



DESIGN DOCUMENT FOR EMPLOYEE MANAGEMENT SYSTEM

Employee Management System Software application



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Problem statement

- The Employment Management System (EEMS) is intended to be the single source of truth for all resource and operational tracking within the company. Your first task is to read this narrative carefully and extract the four core domain entities required to build the system. The company is structured into distinct groups, each managed by a leadership team. These organizational units (departments) have a specific name, a fixed location (building or city), and an annual budget.

These units are responsible for hosting and executing multiple operational tasks. Every person working at the company is considered part of the workforce. For each individual, the system must record their full name, current title (e.g., Manager, Specialist), hire date, and current salary. Crucially, every single member of the workforce must be permanently assigned to one, and only one, organizational unit. Operational tasks are structured as defined projects. Each project must have a name, detailed description, specific start and end dates, and a defined budget. Projects can be in various stages, tracked by their status (e.g., 'Active', 'Completed').

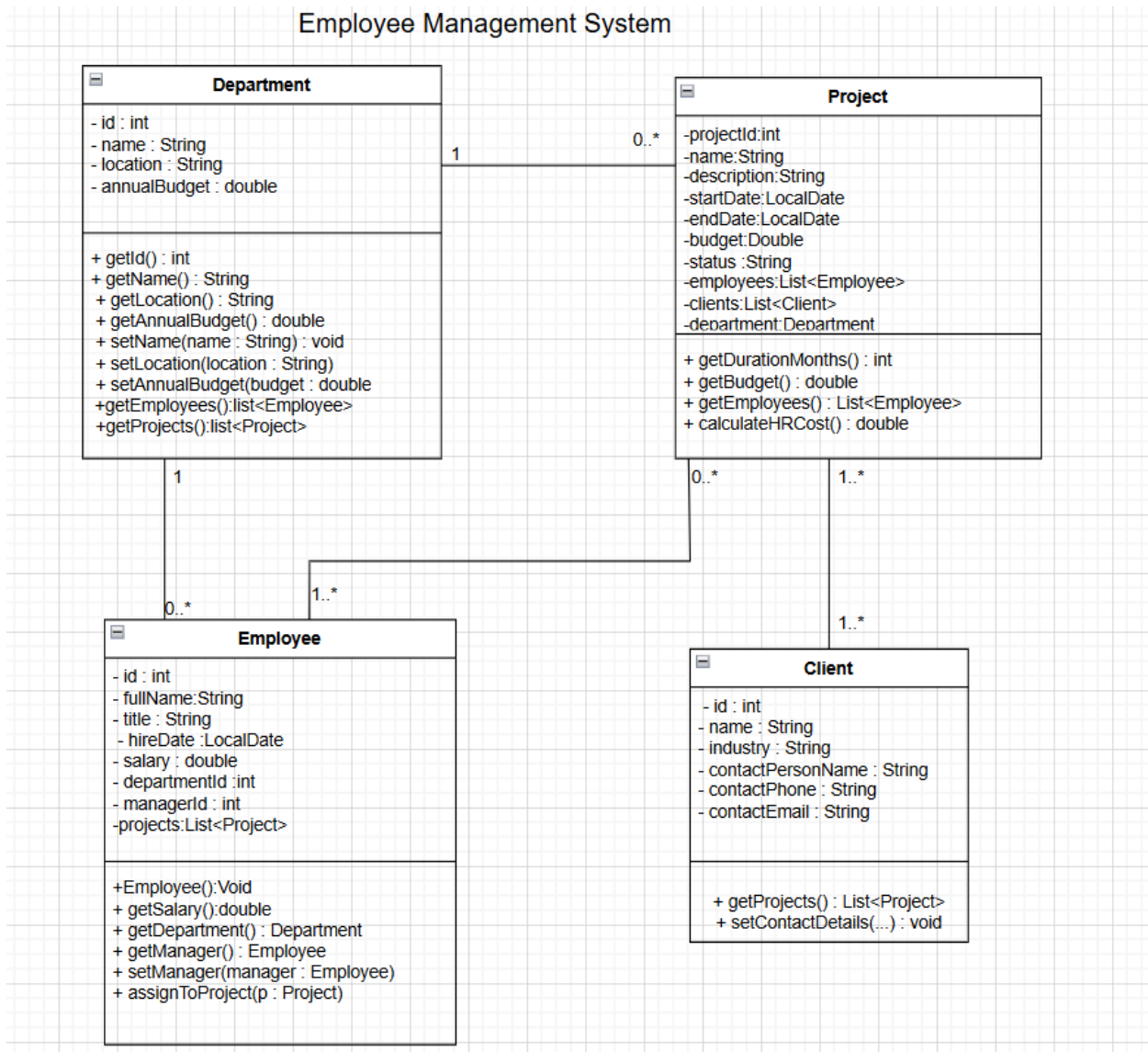
These projects are typically managed or hosted collaboratively by multiple organizational units. Furthermore, the workforce works on these projects. Any single person can be assigned to multiple projects simultaneously, and naturally, any single project will have many people working on it.

The system must track the percentage of time a worker is allocated to each specific project. Finally, the company tracks external relationships via a registry of associated clients. Each client is defined by their name, the industry they operate in, and the primary contact person's name and contact information (phone and email). A single project can be tied to several different clients, and conversely, a single client may sponsor or be associated with numerous different projects across the company. The system must maintain these links accurately. The system must ensure full CRUD (Create, Read, Update, Delete) functionality for all extracted domain entities and maintain strict referential integrity based on the relationships described above

Domain class description

Class Name	Attribute	Data Type	Justification
Client	id	int	Unique identifier for each client
	name	String	Name of the client organization
	industry	String	Industry sector the client belongs to
	contactPersonName	String	Primary contact person for communication
	contactPhone	String	Phone number of the contact person
	contactEmail	String	Email address of the contact person
Department	id	int	Unique identifier for each department
	name	String	Name of the department
	location	String	Physical location of the department
	annualBudget	double	Annual financial budget allocated to the department
Employee	id	int	Unique identifier for each employee
	fullName	String	Full name of the employee
	title	String	Job title or position of the employee
	hireDate	Date	Date when the employee was hired
	salary	double	Salary of the employee
	departmentId	int	ID of the department the employee belongs to
	managerId	Integer	ID of the employee's manager (nullable)
Project	id	int	Unique identifier for each project
	name	String	Name of the project
	description	String	Detailed description of the project
	startDate	Date	Start date of the project
	endDate	Date	End date of the project
	budget	double	Budget allocated to the project
	status	String	Current status of the project (e.g., Active, Completed, On Hold)
DepartmentProject	departmentId	int	Links a department to a project
	projectId	int	Links a project to a department
EmployeeProject	employeeId	int	Links an employee to a project
	projectId	int	Links a project to an employee
	allocationPercent	double	Percentage of time the employee is allocated to the project
ProjectClient	projectId	int	Links a project to a client
	clientId	int	Links a client to a project

Class diagram



User requirements:

Functional requirements:

- Calculate project HR cost
- Generate department project reports
- Identify high-value clients
- Perform employee transfer transactions
- Support full CRUD for employees and projects

Nonfunctional requirements:

- Reliability and accuracy
- User-friendly interface
- Efficient performance
- Secure data handling
- Maintainable and scalable codebase

Use cases:

Actors list (HR Manager, Admin, Project Manager).

Use case1: Calculate a project HR cost for given project

Precondition: Project must be registered on the system.

Goal: The system should be able to calculate a project HR cost for given project

- System displays menu of operations
- User select the calculate HR cost option
- System ask for Project id
- User enter project id
- System displays results

Use case2: Get department project report

Precondition: Department must be registered on the system

Goal: The system should be able to generate and sort the report for a given department

- System displays menu of operations
- user select generate department report option
- System ask for department id and sorting order details
- User enter department id and sorting order details
- System displays list of active projects in the department in sorted order.

Use case3: High-Value Client Identification

Precondition: Client must be linked to at least one project

Goal: The system should be able to display a list of all Client objects that are linked to one or more projects scheduled to end within the specified number of days from the current date.

- System displays menu of operations
- User select find clients by upcoming project option
- System request for deadline details ie number of days till deadline.
- User enter number of days till deadline details
- System displays list of all clients that are linked to the projects

Use case4: Employee Transfer Transaction

Precondition: Employee must be registered on the system

Goal: The system should be able to perfume employee transfer transaction by updating the employee id and other details when employee is moved to different department.

- System displays menu of operations
- user select Employee Transfer option
- System request for transfer details (ie employee id and department id)
- User enter transfer details
- System updates employee details and displays updated success message.

Use case 5: Create employee use case

Precondition: System must be operational.

Goal: The system should be able to create or add a new employee to the database.

- System displays menu of operations
- User select the create employee option
- System ask for employee details (ie first name, last name etc)
- User enter employee details and press enter button
- System create and add a new employee to the database and display confirmation message.

Use case 6: Read/retrieve employee use case

Precondition: Employee must be registered on the system

Goal: The system should be able to retrieve and display the details of a given employee from the database.

- System displays menu of operations
- User select the retrieve/read employee option
- System ask for employee id.
- User enter employee id and press enter button
- System displays the details of the employee corresponding to the entered id.

Use case 7: Update/Modify employee use case

Precondition: Employee must be registered on the system

Goal: The system should be able to update or modify the details of a given employee.

- System displays menu of operations
- User select the update/modify employee option
- System ask for employee id
- User enter employee id and press enter button
- System displays the details of the corresponding employee in editable formate and ask user to modify.
- User modify employee details and press the enter button
- System modifies employee details and displays confirmation message.

Use case 8: Delete employee use case

Precondition: Employee must be registered on the system

Goal: The system should be able to delete or remove a given employee from the database.

- System displays menu of operations
- User select the delete/remove employee option
- System ask for employee id
- User enter employee id and press enter button
- System remove all details of the corresponding employee from the database and display a confirmation message..

Use case 9: Create project use case

Precondition: System must be operational.

Goal: The system should be able to create or add a new employee to the database.

- System displays menu of operations
- User select the create project option
- System asks for project details.
- User enter project details and press enter button
- System create and add a new project to the database and display confirmation message.

Use case 10: Read/retrieve project use case

Precondition: Project must be registered on the system

Goal: The system should be able to retrieve and display the details of a given project from the database.

- System displays menu of operations
- User select the retrieve/read project option
- System ask for project id.
- User enter project id and press enter button
- System displays the details of the project corresponding to the entered id.

Use case 11: Update/Modify project use case

Precondition: Project must be registered on the system

Goal: The system should be able to update or modify the details of a given project.

- System displays menu of operations
- User select the update/modify project option
- System ask for project id
- User enter project id and press enter button
- System displays the details of the corresponding project in editable formate and ask user to modify.
- User modify project details and press the enter button
- System modifies project details and displays confirmation message.

Use case 12: Delete project use case

Precondition: Project must be registered on the system

Goal: The system should be able to delete or remove a given project from the database.

- System displays menu of operations
- User select the delete/remove project option
- System ask for project id
- User enter project id and press enter button
- System remove all details of the corresponding project from the database and display a confirmation message..

Use case 13: Create client use case

Precondition: System must be operational.

Goal: The system should be able to create or add a new client to the database.

- System displays menu of operations
- User select the create client option
- System asks for client details.
- User enter client details and press enter button
- System create and add a new client to the database and display confirmation message.

Use case 14: Read/retrieve client use case

Precondition: Client must be registered on the system

Goal: The system should be able to retrieve and display the details of a given client from the database.

- System displays menu of operations
- User select the retrieve/read client option
- System ask for client id.
- User enter client id and press enter button
- System displays the details of the client corresponding to the entered id.

Use case 15: Update/Modify client use case

Precondition: Client must be registered on the system

Goal: The system should be able to update or modify the details of a given client.

- System displays menu of operations
- User select the update/modify client option
- System ask for client id
- User enter client id and press enter button
- System displays the details of the corresponding client in editable formate and ask user to modify.
- User modify client details and press the enter button
- System modifies client details and displays confirmation message.

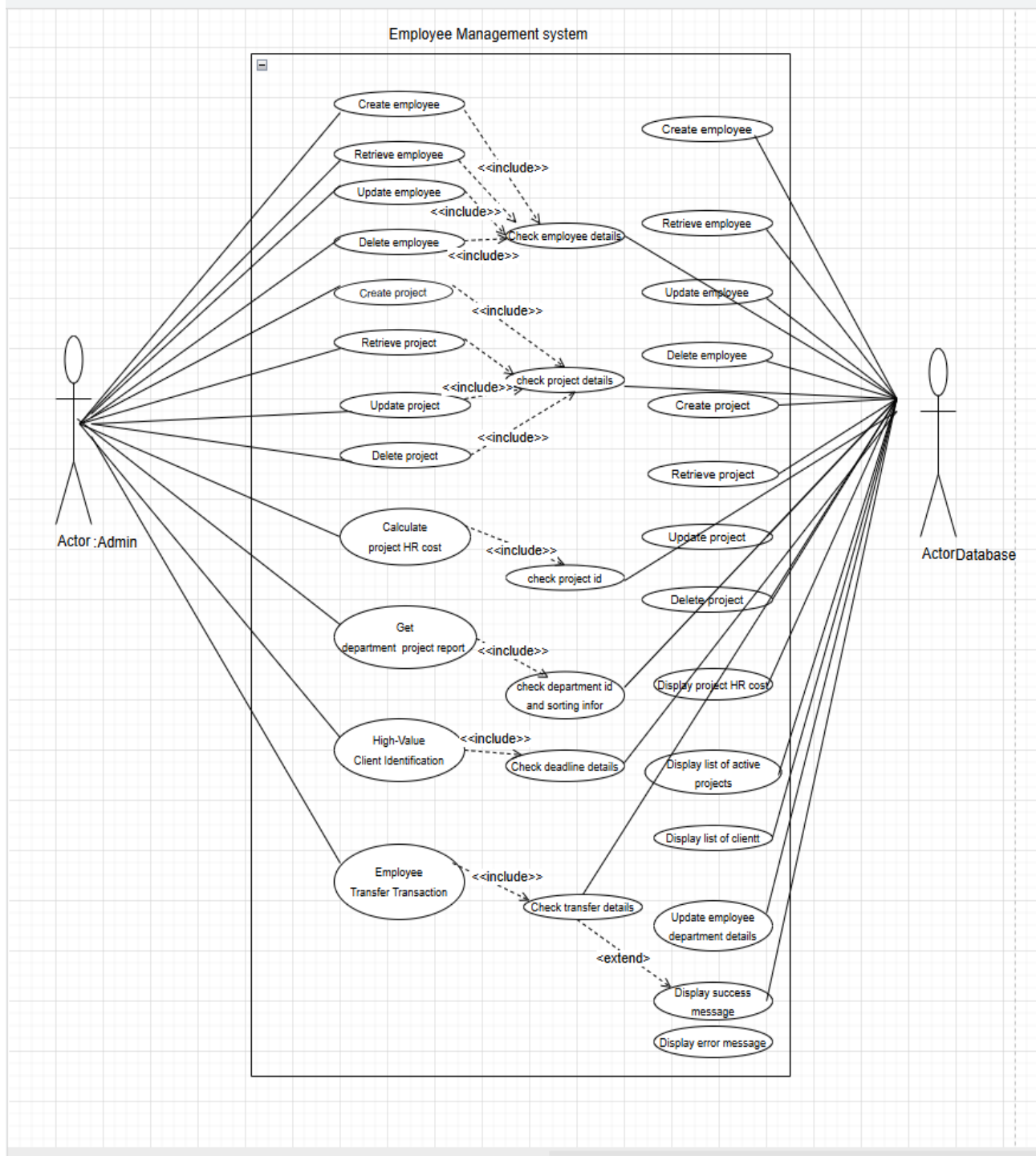
Use case 16: Delete project use case

Precondition: Client must be registered on the system

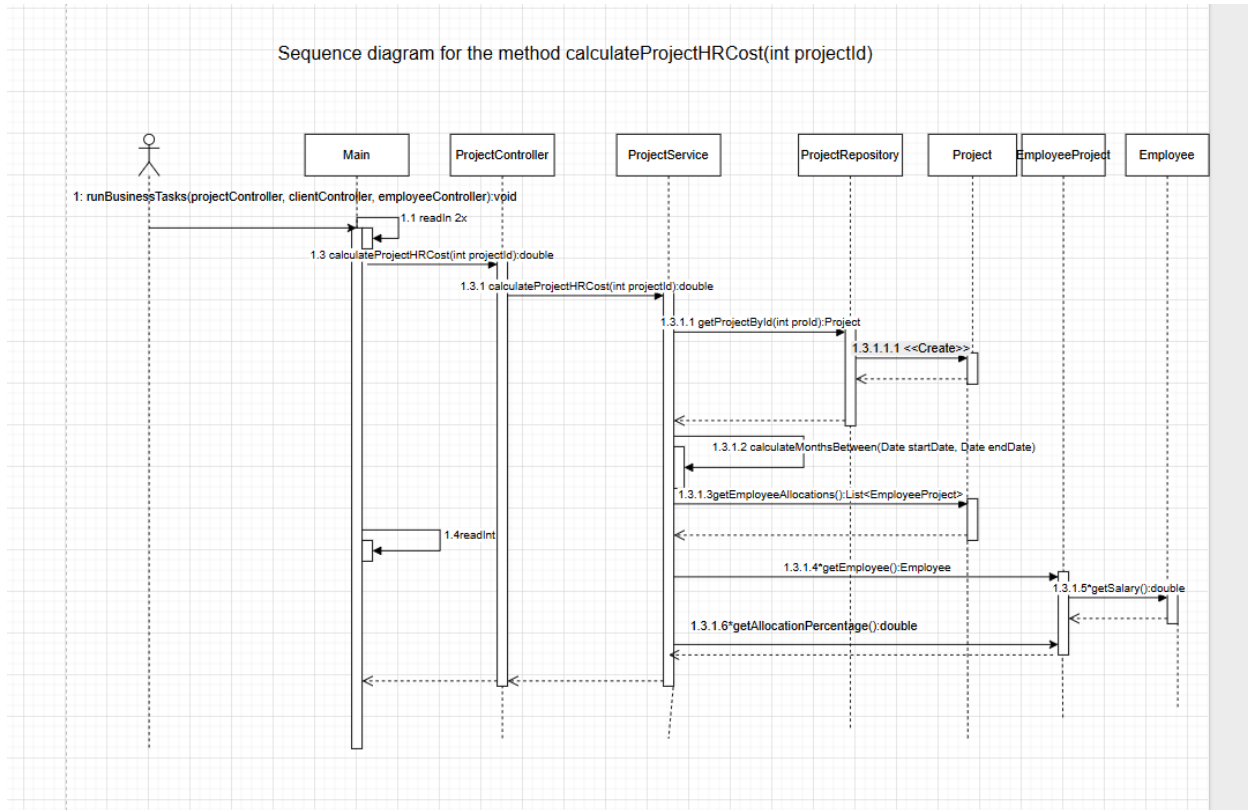
Goal: The system should be able to delete or remove a given client from the database.

- System displays menu of operations
- User select the delete/remove client option
- System ask for client id
- User enter client id and press enter button
- System remove all details of the corresponding client from the database and display a confirmation message..

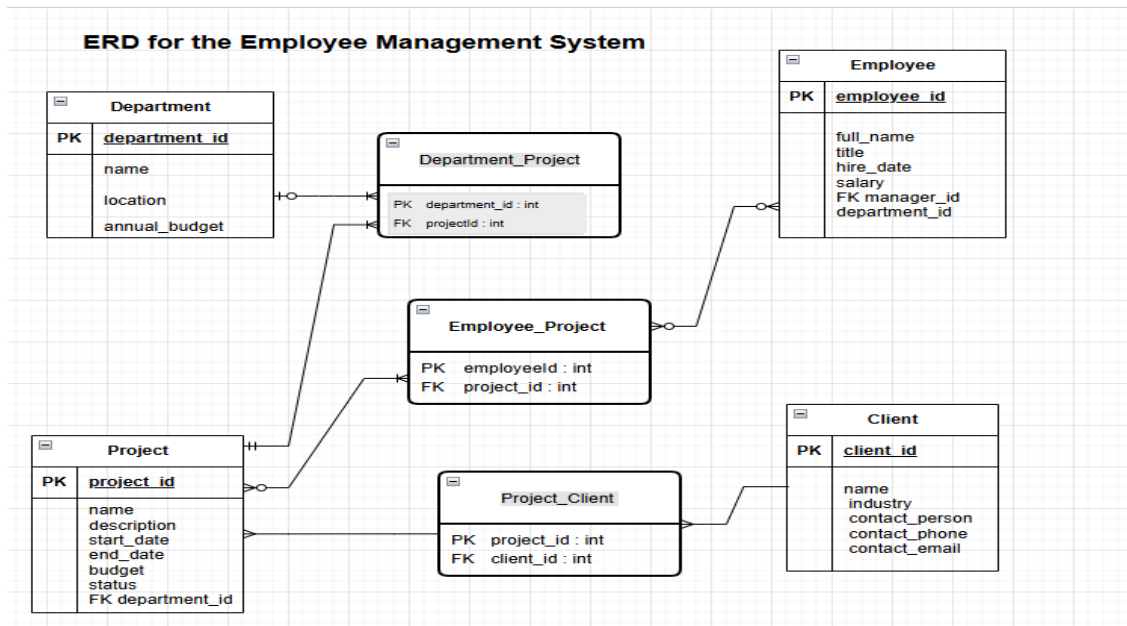
Use case diagram:



Sequence Diagram



Database ERM



Software Implementation

Folder

```

    1 import controller.ClientController;
    2 import controller.EmployeeController;
    3 import controller.ProjectController;
    4 import util.DBConnector;
    5 import java.util.Scanner;
    6
    7 public class Main {
    8
    9     public static void main(String[] args) {
    10         Scanner scanner = new Scanner(System.in);
    11
    12         try {
    13             ProjectController projectController = new ProjectController();
    14             ClientController clientController = new ClientController();
    15             EmployeeController employeeController = new EmployeeController();
    16             while (true) {
    17                 System.out.println("EMPLOYEE MANAGEMENT SYSTEM(ENR)");
    18                 System.out.println("1. Department Management (ENR)");
    19                 System.out.println("2. Employee Management (ENR)");
    20                 System.out.println("3. Client Management (ENR)");
    21                 System.out.println("4. Project Management (ENR)");
    22                 System.out.println("5. Run Business Logic Tests");
    23                 System.out.println("6. Department Project (ENR)");
    24                 System.out.println("7. Employee Project (ENR)");
    25                 System.out.println("8. Project Client (ENR)");
    26                 System.out.println("9. Exit");
    27                 System.out.print("Enter your choice: ");
    28
    29                 int choice = Integer.parseInt(scanner.nextLine());
    30
    31                 switch (choice) {
    32                     case 1 -> DepartmentCLI.main(scanner.nextLine());
    33                     case 2 -> EmployeeCLI.main(scanner.nextLine());
    34                     case 3 -> ClientCLI.main(scanner.nextLine());
    35                     case 4 -> ProjectCLI.main(scanner.nextLine());
    36
    37                 }
    38             }
    39         } catch (Exception e) {
    40             System.out.println("Error: " + e.getMessage());
    41         }
    42     }
    43 }
    
```

```
Project - Version control - Current File - Main.java - Project21.java
Run - Main
EMPLOYMENT MANAGEMENT SYSTEM(EMS)
1. Department Management (CRUD)
2. Employee Management (CRUD)
3. Client Management (CRUD)
4. Project Management (CRUD)
5. Run Business Logic Tests
6. Department Project (CRUD)
7. Employee Project (CRUD)
8. Project Client (CRUD)
9. Exit
Enter your choice: 2

DEPARTMENT CRUD
1. Add Department
2. View All Departments
3. View Department by ID
4. Update Department
5. Delete Department
6. Go back
Choose an option: 3
Enter Department ID: 1
Department: Engineering, Location: New York

DEPARTMENT CRUD
1. Add Department
2. View All Departments
3. View Department by ID
4. Update Department
5. Delete Department
6. Go back
Choose an option: 1
```

```
Project - Version control - Current File - Main.java - Project21.java
Run - Main
Enter Department ID: 1
Department: Engineering, Location: New York

DEPARTMENT CRUD
1. Add Department
2. View All Departments
3. View Department by ID
4. Update Department
5. Delete Department
6. Go back
Choose an option: 4
Exiting from Department CRUD... Goodbye!

EMPLOYMENT MANAGEMENT SYSTEM(EMS)
1. Department Management (CRUD)
2. Employee Management (CRUD)
3. Client Management (CRUD)
4. Project Management (CRUD)
5. Run Business Logic Tests
6. Department Project (CRUD)
7. Employee Project (CRUD)
8. Project Client (CRUD)
9. Exit
Enter your choice: 5

BUSINESS LOGIC TESTS
1. Calculate Project HR Cost
2. Get Projects by Department
3. Find Clients by upcoming Project Deadline
4. Transfer Employees to Another Department
5. Return to Main Menu
Enter your choice:
```